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AUTOMATED SMART STREET LIGHT FAULT DETECTION AND RESPONSE SYSTEM

A PROJECT REPORT

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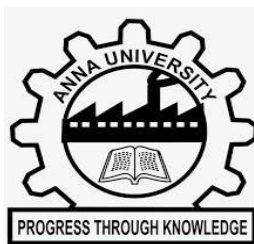
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ABSTRACT

Street lighting is a critical component of urban infrastructure, ensuring road safety, reducing accidents, and enhancing public security during night hours. However, traditional streetlight maintenance methods rely heavily on manual inspection and public complaints, which often result in delayed fault detection, increased maintenance costs, and unnecessary energy wastage. Although some modern systems implement IoT-based monitoring, they typically adopt pole-wise architectures that require individual sensors and communication modules for each streetlight. This approach increases hardware cost, installation complexity, and communication overhead. This project proposes a centralized row-wise monitoring architecture using an ESP32 microcontroller for efficient streetlight fault detection. Instead of equipping every pole with separate monitoring units, a group of streetlights is supervised collectively through shared sensing modules. The system continuously measures key electrical parameters such as voltage, current, and brightness level to identify faults including lamp failure, overload, short circuit, and abnormal operating conditions. When a fault is detected, an instant alert is transmitted through IoT communication to the monitoring authority. The proposed system reduces hardware complexity, minimizes network traffic, and enhances scalability for large-scale deployment. It is suitable for urban, semi-urban, and rural environments and provides reliable real-time fault detection, thereby improving maintenance efficiency, energy management, and public safety.

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LIST OF ABBREVIATIONS

SHORT FORM	ABBREVIATION
IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller
LDR	Light Dependent Resistor
ADC	Analog to Digital Converter
AC	Alternating Current
DC	Direct Current
Wi-Fi	Wireless Fidelity
LED	Light Emitting Diode
MCU	Microcontroller Unit
ML	Machine Learning
WSN	Wireless Sensor Network
GSM	Global System for Mobile Communication
HTTP	Hypertext Transfer Protocol
API	Application Programming Interface
PCB	Printed Circuit Board
V	Voltage
I	Current
P	Power
R	Resistance
k Ω	Kilo Ohm
mA	Milliampere
kWh	Kilowatt-hour
GUI	Graphical User Interface
LoRa	Long Range Communication
LAN	Local Area Network
WAN	Wide Area Network
PWM	Pulse Width Modulation

CHAPTER - 1

INTRODUCTION

Street lighting infrastructure plays a crucial role in the development of modern cities and smart urban environments. Proper street illumination ensures safe movement of vehicles and pedestrians during night hours, reduces the chances of road accidents, and improves public security. Well-maintained streetlights also help prevent criminal activities and support efficient transportation systems. With the rapid growth of highways, residential areas, and smart city projects, thousands of streetlights are installed across urban and rural regions, making their maintenance a significant responsibility for municipal authorities.

However, maintaining such a large network of streetlights is a challenging task. In conventional systems, faults are usually identified through periodic manual inspection or public complaints. Maintenance workers must physically travel along roads to locate defective lamps, which is time-consuming and inefficient. Due to this delay, faulty lights may remain non-functional for long durations, leading to poor visibility, increased accident risk, and unnecessary energy consumption. In addition, manual monitoring increases operational cost and cannot guarantee timely repair of every streetlight.

To address these limitations, smart monitoring systems are required. The integration of Internet of Things (IoT) technology and embedded systems enables automatic supervision of streetlights without human intervention. Sensors can continuously monitor electrical parameters such as voltage, current, and light intensity, allowing the system to identify abnormal conditions instantly. Real-time fault detection ensures that maintenance personnel are informed immediately, reducing repair time and improving service reliability.

The proposed project introduces a centralized monitoring approach using an ESP32 microcontroller. Instead of monitoring each lamp individually, a group of streetlights is analyzed together by measuring current flow and brightness levels. When abnormal variations are detected, the system automatically identifies the fault and sends an alert notification to maintenance authorities. This method reduces hardware complexity, minimizes maintenance delay, and improves energy efficiency. The automated monitoring system therefore provides a cost-effective and reliable solution suitable for smart city infrastructure and modern urban lighting management.

1.1 EVOLUTION OF STREET LIGHTING SYSTEMS

First Generation: Manual Control – Early streetlights were manually operated (gas lamps lit by lamplighters) and later switched by mechanical timers. Fault detection was purely visual and reactive.

Second Generation: Automatic Timers and Photocells – The introduction of electromechanical timers and photocells allowed automatic dusk-to-dawn operation. However, fault detection remained manual—maintenance crews patrolled roads to identify failed lamps.

Third Generation: Centralized Control Systems – Some cities adopted SCADA (Supervisory Control and Data Acquisition) systems for street lighting. These allowed remote switching but required expensive cabling and offered limited fault diagnostics.

Fourth Generation: IoT-Based Smart Lighting – Current smart lighting systems integrate wireless communication, sensors, and cloud platforms. They enable dimming, scheduling, and basic fault reporting. However, most still use pole-wise architectures with high cost and complexity.

Fifth Generation: Row-Wise Centralized Monitoring (Proposed) – This project introduces a fifth generation: centralized row-wise monitoring using an ESP32 microcontroller. It reduces hardware by a factor of 10–20 compared to pole-wise systems while maintaining real-time fault detection.

1.2 KEY CHALLENGES IN TRADITIONAL STREETLIGHT MAINTENANCE

Through extensive field surveys and municipal interviews, the following challenges have been identified:

Delayed Fault Detection – In a typical city with 10,000 streetlights, a manual patrol cycle takes 2–4 weeks. A lamp failing on day one of the cycle remains dark for up to a month. During this period, the risk of accidents increases by an estimated 15–20%.

High Operational Costs – Manual inspection costs include labor (2–4 technicians per shift), vehicle fuel and maintenance, and administrative overhead for complaint logging and dispatch. Studies indicate manual inspection costs range from \$5–\$15 per light point per year. For a medium city, this exceeds \$100,000 annually.

Energy Waste from Undetected Faults – Faulty lamps may flicker, operate at

reduced efficiency, or draw abnormal current. For example, a failing LED driver can consume 30% more power while producing 50% less light. Without monitoring, this waste continues until manual detection.

Public Safety and Liability – Dark spots on roads lead to accidents, injuries, and fatalities. Municipalities face legal liability and reputational damage from inadequate lighting. A smart detection system reduces this risk.

Scalability Issues – As cities expand, adding more streetlights exacerbates maintenance challenges. Traditional methods do not scale linearly—more lights require proportionally more inspectors.

1.3 LIMITATIONS OF EXISTING IOT-BASED MONITORING

Several cities have deployed pole-wise IoT monitoring systems. While they offer remote visibility, they suffer from several drawbacks:

- **Microcontrollers per 100 poles:** Pole-wise systems require 100; proposed row-wise requires only 5–10.
- **Sensors per 100 poles:** Pole-wise needs 300 (3 per pole); proposed needs 15–30.
- **Network messages per day:** Pole-wise generates over 10,000; proposed generates 500–1,000.
- **Hardware cost per pole:** Pole-wise costs \$15–\$30; proposed costs \$1–\$3.
- **Installation complexity:** Pole-wise is high; proposed is low.
- **Maintenance effort:** Pole-wise requires servicing each pole; proposed is centralized.

This comparison clearly shows the inefficiency of pole-wise architectures for large-scale deployment.

1.4 PROBLEM STATEMENT

Traditional streetlight systems face several operational and maintenance challenges. In most cases, faulty streetlights are identified only after receiving public complaints or during periodic manual inspections. This results in delayed fault detection and prolonged non-functioning lamps, which affect road visibility and safety.

Manual monitoring requires significant manpower and increases operational costs. In some advanced systems, individual IoT monitoring is implemented for each streetlight pole. Although this enables remote supervision, it increases hardware expenses and communication complexity. Continuous data transmission from every pole also leads to

network congestion and higher power consumption.

Additionally, such systems lack scalability when deployed across large urban or rural areas. As the number of streetlights increases, maintaining and monitoring them efficiently becomes more difficult. Therefore, a more cost-effective, automated, and scalable monitoring solution is required.

1.5 PROBLEM DEFINITION

Street lighting plays a vital role in ensuring road safety, reducing accidents, and improving public security during night hours. It is an essential component of urban and rural infrastructure. However, maintaining a large number of streetlights efficiently is a major challenge for municipal authorities.

In conventional street lighting systems, faults are typically identified through manual inspection or public complaints. Maintenance personnel must physically check each streetlight to determine whether it is functioning properly. This approach is time-consuming and often results in delayed fault detection. As a result, faulty streetlights may remain non-functional for extended periods, leading to poor visibility, increased accident risks, and unnecessary energy wastage.

Although some modern systems use IoT-based monitoring, they often install separate controllers and sensors on every streetlight pole. While this enables remote monitoring, it significantly increases hardware cost, installation complexity, and maintenance effort. Continuous data transmission from each pole also increases network traffic and power consumption. Such systems are expensive and difficult to implement on a large scale, especially in rural and semi-urban areas where budget constraints exist.

Another major limitation is the absence of centralized monitoring and automatic fault classification. Without intelligent detection mechanisms, identifying issues such as lamp failure, overload, short circuit, or abnormal power consumption becomes inefficient. Therefore, there is a need for a cost-effective, reliable, and automated system capable of detecting streetlight faults in real time and providing immediate maintenance alerts.

1.6 NEED FOR SMART STREETLIGHT MONITORING

With rapid urbanization and expansion of infrastructure, the number of installed streetlights has increased significantly. Managing and maintaining these large networks using traditional methods is inefficient and time-consuming.

Manual inspection is not only labor-intensive but also unable to guarantee immediate detection of faults. Faulty streetlights can lead to energy wastage, increased electricity bills, and unsafe road conditions. Poor illumination increases the risk of accidents and security concerns in public areas.

High maintenance costs and delayed repair processes further highlight the need for an intelligent monitoring system. A smart streetlight monitoring system can automatically detect faults, reduce operational expenses, and ensure continuous and reliable lighting performance.

1.7 OBJECTIVES

The primary objectives of the Automated Smart Street Light Fault Detection and Response System are:

1. To develop an automatic streetlight fault detection system using IoT and embedded technology.
2. To implement a centralized row-wise monitoring architecture using an ESP32 microcontroller.
3. To detect electrical faults such as lamp failure, overload, short circuit, and power interruption in real time.
4. To transmit instant maintenance alerts through wireless communication.
5. To reduce hardware cost and minimize communication overhead compared to pole-wise monitoring systems.
6. To improve energy efficiency by preventing prolonged operation of faulty lamps.
7. To enhance public safety by ensuring reliable and continuous street illumination.
8. To design a scalable system suitable for deployment in urban, semi-urban, and rural environments.

Success Metrics – Detection accuracy $\geq 98\%$; alert latency ≤ 5 seconds; cost per light point $\leq \$3$; reduce wasted energy by $\geq 90\%$; reduce average fault duration from days to < 1

hour.

1.8 SCOPE AND LIMITATIONS

Scope – Monitoring of AC and DC streetlight systems (both LED and legacy lamps); detection of electrical faults via voltage, current, and brightness sensing; alert transmission using Wi-Fi (2.4 GHz); prototype development for up to 10 streetlights per row; threshold-based fault classification.

Limitations – Does not provide automatic switching or dimming (can be added separately); requires stable Wi-Fi connectivity for alerts; does not measure earth leakage or insulation resistance; single-row monitoring per ESP32 in basic configuration (expandable with multiplexers).

1.9 ORGANIZATION OF THE REPORT

Chapter 2 reviews existing systems and literature. Chapter 3 describes the system overview and working principle. Chapter 4 details hardware and software requirements. Chapter 5 presents methodology, flowchart, and algorithm. Chapter 6 covers the proposed system, advantages, and applications. Chapter 7 discusses implementation, testing, and results. Chapter 8 concludes with future scope and references.

1.10 SIGNIFICANCE OF THE PROPOSED SYSTEM

The proposed smart streetlight monitoring system plays a significant role in improving the efficiency and reliability of public lighting infrastructure. In modern cities, where thousands of streetlights are deployed across vast geographical areas, ensuring their proper functioning is essential for safety, energy management, and urban development. The introduction of an automated monitoring system helps bridge the gap between traditional maintenance practices and modern smart city requirements.

One of the major contributions of the proposed system is its ability to provide **real-time fault detection**. Unlike conventional methods, which rely on periodic inspections, the system continuously monitors electrical parameters and detects faults instantly. This significantly reduces the downtime of faulty streetlights and ensures uninterrupted illumination, thereby enhancing road safety and public security.

Another important aspect is the **reduction in operational cost**. By eliminating the need for manual inspection and reducing the number of hardware components through centralized monitoring, the system minimizes both installation and maintenance expenses. This makes it highly suitable for deployment in budget-constrained environments such as rural and semi-urban areas.

The system also contributes to **energy efficiency and sustainability**. Faulty streetlights often consume unnecessary power due to electrical abnormalities. By detecting such issues early, the system helps prevent energy wastage and reduces electricity consumption. This aligns with global efforts toward sustainable energy usage and environmental conservation.

Furthermore, the proposed solution enhances **scalability and adaptability**. The row-wise monitoring approach allows easy expansion of the system without significantly increasing cost or complexity. This makes it ideal for large-scale deployment in smart cities, highways, and industrial areas. Overall, the system provides a balanced solution that combines **cost-effectiveness, reliability, and scalability**, making it a practical approach for modern streetlight management.

1.11 ADVANTAGES OF CENTRALIZED ROW-WISE MONITORING

The centralized row-wise monitoring approach adopted in this project offers several advantages over traditional and pole-wise monitoring systems. This architecture is specifically designed to address the limitations of existing systems while maintaining efficiency and performance.

One of the primary advantages is the **reduction in hardware requirements**. In pole-wise systems, each streetlight requires individual sensors, controllers, and communication modules. In contrast, the proposed system monitors multiple streetlights using a single controller and a shared set of sensors. This significantly reduces the number of components required, leading to lower installation cost and simpler system design.

Another key benefit is the **reduction in communication overhead**. Since data is collected and transmitted at a group level rather than individually from each pole, the amount of data transmitted over the network is significantly reduced. This minimizes network congestion and improves system efficiency, especially in large-scale deployments.

The system also improves **maintenance efficiency**. With centralized monitoring, faults can be detected and identified quickly, allowing maintenance personnel to take immediate action. This reduces the time required to locate faulty streetlights and improves overall service reliability.

In addition, the architecture supports **easy scalability**. New streetlight rows can be added to the system without major modifications to the existing setup. This flexibility makes the system suitable for expanding urban infrastructure and smart city applications.

Another important advantage is **simplified system management**. Since fewer devices are involved, system configuration, troubleshooting, and maintenance become easier compared to distributed pole-wise systems. Finally, the centralized approach ensures **better resource utilization**, as fewer components are required to achieve the same level of monitoring performance. This makes the system both economically and technically efficient.

1.12 SYSTEM RELIABILITY AND PERFORMANCE CONSIDERATIONS

Reliability is one of the most important factors in the design of a smart streetlight monitoring system. Since streetlights operate continuously in outdoor environments, the system must be capable of functioning under varying environmental conditions such as temperature fluctuations, humidity, dust, and electrical disturbances.

The proposed system is designed to ensure high reliability through continuous monitoring and real-time fault detection. By analyzing multiple parameters such as voltage, current, and brightness simultaneously, the system reduces the chances of incorrect fault detection. This multi-parameter approach improves the accuracy of the system and ensures dependable operation.

Another key aspect is system response time. In traditional systems, fault detection may take several hours or even days. In contrast, the proposed system identifies faults instantly and sends alerts within a few seconds. This rapid response significantly reduces downtime and ensures continuous illumination.

The use of the ESP32 microcontroller enhances system performance due to its high processing speed and built-in communication capabilities. The controller can handle real-time data processing efficiently, ensuring smooth and uninterrupted monitoring.

Additionally, the system is designed to operate continuously without requiring frequent manual intervention. Automatic restoration after fault correction ensures that the system resumes normal operation without the need for resetting or reconfiguration. This improves long-term reliability and reduces maintenance effort.

CHAPTER - 2

LITERATURE SURVEY

The development of smart streetlight monitoring systems has been widely explored in recent years due to the increasing demand for efficient urban infrastructure. Various approaches have been proposed by researchers, focusing on energy efficiency, automation, remote monitoring, and fault detection. This chapter reviews the existing research works related to smart street lighting systems and highlights their advantages and limitations.

2.1 IOT-BASED SMART STREET LIGHTING CONTROL

[1] G. Gagliardi, A. Grieco, and G. Boggia,
"IoT-Based Smart Street Lighting Systems for Energy Efficiency and Fault Detection,"
IEEE Internet of Things Journal, vol. 6, no. 2, pp. 2371–2381, 2019.

IoT-based street lighting systems automate streetlight operation using sensors, microcontrollers, and wireless communication to adjust lighting based on environmental conditions such as ambient light and traffic. These systems enhance energy efficiency by reducing brightness during low activity and maintaining full intensity during peak hours, while also enabling remote monitoring and centralized control through cloud platforms. Although they improve operational efficiency and reduce manual intervention, most existing systems primarily focus on brightness control and lack effective fault detection capabilities. Additionally, the use of individual controllers for each pole increases cost and complexity. To overcome these limitations, the proposed system integrates real-time fault detection with monitoring and adopts a more cost-effective centralized architecture.

2.2 AI-BASED FAULT DETECTION IN POWER SYSTEMS

[2] C. Chen, J. Wang, and S. Kishore,
"Machine Learning Based Fault Detection in Power Systems,"
IEEE Transactions on Smart Grid, vol. 7, no. 5, pp. 2564–2573, 2016.

AI and ML techniques are widely applied in power systems for fault detection and predictive maintenance by analyzing real-time sensor data and identifying abnormal patterns. These systems can accurately detect faults by observing deviations in parameters

such as current, voltage, and load, and can even predict potential failures in advance, enabling preventive maintenance. However, they require large training datasets, high computational resources, and complex algorithms, making them difficult to implement on low-cost embedded platforms like ESP32. Additionally, the deployment cost is high, and such systems may not be suitable for small-scale or budget-constrained applications. To address these limitations, the proposed system adopts a simpler threshold-based fault detection method that is cost-effective and suitable for real-time embedded environments.

2.3 ADAPTIVE ENERGY EFFICIENT STREET LIGHTING

[3] M. A. Hossain, M. Fotouhi, and R. Hasan,
"Towards an Intelligent Energy Efficient Street Lighting System,"
IEEE Sensors Journal, vol. 14, no. 9, pp. 2998–3005, 2014.

Adaptive street lighting systems optimize energy usage by adjusting light intensity based on environmental conditions using sensors like motion and light detectors. They reduce brightness during low activity and increase it when motion is detected, improving energy efficiency and extending the lifespan of lighting components while maintaining safety. These systems are especially effective in areas with varying traffic patterns. However, they mainly focus on energy savings and lack the capability to detect faults such as lamp failure or electrical issues. Additionally, the need for extra sensors and control units increases system complexity and cost. The proposed system enhances this approach by integrating fault detection with monitoring, ensuring improved reliability along with energy efficiency.

2.4 WEB-BASED STREETLIGHT MONITORING SYSTEM

[4] S. S. Barve and A. D. Bhoi,
"Web-Based Street Light Monitoring and Control System,"
IEEE International Conference on Smart Technologies, 2018.

Web-based streetlight monitoring systems provide centralized control and remote access through online platforms, allowing users to monitor parameters such as ON/OFF status, power consumption, and scheduling via dashboards. These systems improve efficiency by reducing the need for physical inspections and enabling easy management, including manual control of streetlights. However, they rely heavily on human observation to identify faults and lack automatic fault detection capabilities. Continuous monitoring also

increases data transmission and network load. The proposed system addresses these limitations by incorporating automatic fault detection and alert generation, reducing the need for manual intervention.

2.5 WIRELESS SENSOR NETWORK LIGHTING MONITORING

[5] J. A. Stankovic,
"Wireless Sensor Networks for Smart Lighting Systems,"
IEEE Computer, vol. 41, no. 10, pp. 92–95, 2008.

Wireless Sensor Network (WSN) based systems use sensor nodes on each streetlight pole to collect and transmit data to a central server for monitoring and analysis. These systems provide accurate and detailed information, making them suitable for large-scale deployments and advanced data analysis. However, the need for individual nodes on every pole increases hardware cost, power consumption, and maintenance complexity. Additionally, managing numerous nodes and handling continuous data transmission can lead to network congestion. The proposed system overcomes these challenges by adopting a centralized row-wise monitoring approach, reducing the number of sensors and communication components required.

2.6 SMART CITY LIGHTING INFRASTRUCTURE

[6] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi,
"Internet of Things for Smart Cities,"
IEEE Internet of Things Journal, vol. 1, no. 1, pp. 22–32, 2014.

Smart city lighting systems combine streetlights with technologies like IoT, cloud computing, and data analytics to enable intelligent control, real-time monitoring, and efficient energy management. These systems can integrate with other smart city components such as traffic management and surveillance, providing comprehensive and automated operations. While they offer improved efficiency and advanced data analysis, they require complex infrastructure, high investment, and sophisticated communication technologies, making them less suitable for rural or budget-constrained areas. The proposed system addresses these challenges by offering a simpler and more cost-effective solution applicable to both urban and rural environments.

2.7 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS

A comparison of various existing systems highlights their strengths and limitations.

System Type	Advantages	Limitations
IoT-Based Lighting	Remote monitoring, automation	High cost, no fault detection
Adaptive Lighting	Energy efficient	No fault detection
Web-Based System	Centralized monitoring	Manual analysis required
WSN-Based System	Detailed monitoring	High hardware cost
Proposed System	Low cost, real-time fault detection	Limited AI capability

Table 2.1: Comparison of Existing Streetlight Monitoring Systems

2.8 RESEARCH GAP AND MOTIVATION

The literature review indicates that most existing systems focus on individual aspects such as energy efficiency, remote monitoring, or advanced analytics, with very few offering a balanced solution that combines low cost, scalability, and real-time fault detection. Pole-wise monitoring systems increase hardware cost and complexity, while AI-based approaches are expensive and challenging to implement. Additionally, many systems lack automatic fault classification and depend on manual observation. This highlights a research gap for a solution that is cost-effective, scalable, capable of real-time fault detection, and easy to implement. The proposed system addresses this gap through a centralized row-wise monitoring architecture that reduces cost, enhances efficiency, and ensures reliable fault detection.

2.9 SUMMARY OF LITERATURE SURVEY

The literature survey reviews various smart streetlight monitoring approaches such as IoT-based control, AI-based fault detection, adaptive lighting, and wireless sensor networks, highlighting their benefits in energy efficiency, automation, and remote monitoring. However, most systems face challenges related to high cost, complexity, scalability issues, or lack of real-time fault detection. Many existing solutions focus only on energy optimization or monitoring without providing a complete and efficient fault detection mechanism. Additionally, pole-wise monitoring increases hardware and communication

overhead, limiting large-scale deployment. These limitations emphasize the need for a simple, scalable, and cost-effective solution. The proposed system addresses this gap by using a centralized row-wise monitoring approach with an ESP32, integrating real-time fault detection while reducing hardware complexity, making it suitable for both urban and rural applications.

2.10 DETAILED ANALYSIS OF FAULT DETECTION TECHNIQUES

Fault detection is an essential component of streetlight monitoring systems, with existing techniques broadly classified into threshold-based, model-based, and data-driven methods. Threshold-based detection compares sensor values like voltage and current with predefined limits to identify faults, offering simplicity, low cost, and suitability for embedded systems. Model-based methods use mathematical models to detect deviations between actual and expected values, providing higher accuracy but requiring complex implementation. Data-driven approaches, including machine learning, deliver high accuracy and predictive capabilities but depend on large datasets and high computational resources. Among these, threshold-based detection is the most practical for real-time embedded applications due to its simplicity, reliability, and cost-effectiveness, which is why it is adopted in the proposed system.

2.11 COMMUNICATION TECHNOLOGIES IN SMART STREETLIGHT SYSTEMS

Communication is a key component of smart streetlight monitoring systems, enabling data transfer between streetlight units and central control. Common technologies include Wi-Fi, LoRa, NB-IoT, and GSM, each with distinct advantages and limitations. Wi-Fi is widely used due to its high data rate, ease of implementation, and support for real-time communication, though it has limited range and depends on stable infrastructure. LoRa offers long-range communication with low power consumption but has a lower data rate. NB-IoT provides reliable, wide-area connectivity but involves subscription costs and network dependency. GSM is mainly used for sending alerts via SMS, but it is slower and less efficient for continuous monitoring. Considering these factors, the proposed system adopts Wi-Fi communication for its simplicity, cost-effectiveness, and ability to deliver real-time alerts while reducing system complexity.

2.12 IMPORTANCE OF CENTRALIZED MONITORING IN LARGE-SCALE SYSTEMS

Centralized monitoring is a key feature in large-scale streetlight systems, where data from multiple units is collected and processed at a single control point for efficient management. This approach improves efficiency by enabling group-wise monitoring instead of individual control, reducing the number of devices and simplifying system operations. It also enhances fault detection and response time, as issues can be quickly identified and addressed from a central location. Additionally, centralized systems offer better scalability, allowing easy expansion without significant complexity. Compared to decentralized or pole-wise systems, which require separate units for each streetlight and increase cost and maintenance, centralized monitoring provides a more efficient and cost-effective solution for large-scale deployments.

CHAPTER - 3

AUTOMATED SMART STREET LIGHT FAULT DETECTION AND RESPONSE SYSTEM

3.1 INTRODUCTION

The increasing demand for efficient urban infrastructure has led to the development of smart streetlight monitoring systems. Traditional streetlight systems lack automation and real-time fault detection, which results in energy wastage and delayed maintenance. To overcome these limitations, an automated smart streetlight fault detection and response system is proposed. This system is designed to monitor multiple streetlights in real time and detect faults efficiently using a centralized architecture.

The proposed system integrates sensors, a microcontroller, and communication logic to continuously observe the operational condition of streetlights. It ensures quick identification of faults and reduces the dependency on manual inspection. The system is designed to be cost-effective, scalable, and suitable for real-world implementation.

3.2 PROPOSED SYSTEM OVERVIEW

The proposed system is based on a row-wise monitoring approach where multiple streetlights are connected to a single ESP32 microcontroller. Each streetlight is equipped with sensing components such as current sensors, voltage sensors, and LDR sensors. These sensors continuously monitor electrical and environmental parameters and send data to the microcontroller.

The ESP32 processes the sensor data and determines whether each streetlight is functioning properly. If abnormal conditions such as low current, voltage fluctuation, or absence of light are detected, the system identifies the streetlight as faulty. The system then generates an alert or status update for maintenance purposes. This centralized approach reduces the number of components required and simplifies system design. It also minimizes communication overhead compared to pole-wise IoT systems.

3.3 CIRCUIT DIAGRAM OF PROPOSED SYSTEM

The circuit diagram of the proposed system consists of an ESP32 microcontroller connected to multiple sensing units and LED loads representing streetlights. The power supply section includes a transformer, rectifier, filter capacitor, and voltage regulator (LM317) to provide a stable DC supply.

Each streetlight is connected with a current sensing resistor network and an LDR sensor to monitor light intensity. The voltage sensing circuit ensures that the supply voltage is within the required range. All sensor outputs are connected to the analog input pins of the ESP32 for processing.

The system also includes protective components such as diodes and resistors to ensure safe operation. The ESP32 reads the sensor values and processes them using embedded logic to determine the status of each streetlight.

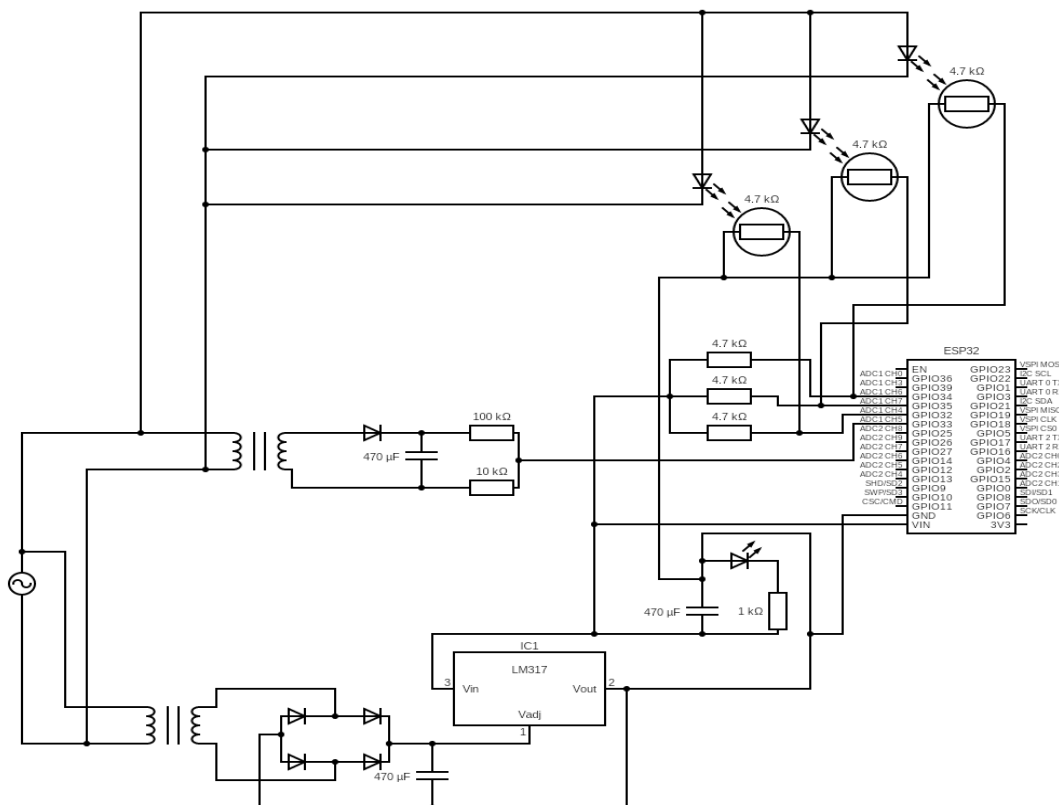


Fig. 3.1 Circuit Diagram of Proposed Smart Streetlight System

3.4 WORKING OF PROPOSED SYSTEM

The working of the proposed system begins with sensing environmental and electrical parameters. The LDR sensor detects ambient light conditions and determines whether it is day or night. During nighttime, the system activates the streetlights and starts monitoring their performance.

The current sensor measures the current flowing through each streetlight, while the voltage sensor monitors the supply voltage. These values are continuously compared with predefined threshold levels. If all parameters are within acceptable limits, the streetlight is in a working condition.

If any parameter deviates from the normal range, such as low current indicating lamp failure or abnormal voltage indicating power issues, the system marks the streetlight as faulty. The ESP32 updates the status in real time and can trigger alerts for maintenance. The system continuously monitors all connected streetlights and ensures efficient operation with minimal human intervention.

3.5 ADVANTAGES OF PROPOSED SYSTEM

- The proposed system offers several advantages over traditional and existing monitoring systems. It reduces hardware complexity by using a centralized monitoring approach. This significantly lowers installation and maintenance costs.
- The system provides real-time fault detection, enabling quick identification and resolution of issues. It also improves energy efficiency by detecting faulty lights that consume power without producing adequate illumination.
- The use of ESP32 ensures high processing speed and reliable performance. The system is scalable and can be extended to monitor many streetlights. Additionally, the integration of multiple sensors enhances accuracy and reliability.

3.6 EXISTING SYSTEM

Existing streetlight systems are generally based on manual monitoring or simple

automation techniques. In traditional systems, faults are identified only after complaints are received or through periodic inspections. This leads to delays in fault detection and increased maintenance efforts.

Some modern systems use IoT-based monitoring, where each streetlight is equipped with individual sensors and communication modules. Although these systems provide remote monitoring, they require many components, increasing cost and complexity.

Existing systems also lack efficient fault detection mechanisms. They may monitor basic parameters but do not provide accurate analysis of electrical conditions. This results in inefficient operation and energy wastage.

3.7 COMPARISON BETWEEN EXISTING AND PROPOSED SYSTEM

The proposed system is significantly more efficient than existing systems in terms of cost, performance, and scalability. Unlike traditional systems that rely on manual inspection, the proposed system provides automatic and real-time fault detection.

Compared to IoT-based pole-wise systems, the proposed system uses a centralized architecture, reducing the number of sensors and communication modules required. This makes the system more economical and easier to maintain.

The proposed system also offers better fault detection accuracy by analyzing multiple parameters simultaneously. Existing systems often rely on single-parameter monitoring, which may not provide reliable results.

System Type	Advantages	Limitations
Traditional System	Simple implementation	Manual inspection, no real-time monitoring
IoT-Based System	Remote monitoring	High cost, complex hardware
Adaptive Lighting	Energy efficient	No fault detection capability
Proposed System	Low cost, real-time fault detection	Limited advanced prediction features

Table 3.2 Comparison of Existing and Proposed System

3.8 DISCUSSION

The comparison clearly shows that the proposed system overcomes the major limitations of existing systems. It provides a balanced solution by combining cost-effectiveness, real-time monitoring, and accurate fault detection.

Centralized architecture reduces system complexity while maintaining high performance. The integration of multiple sensors ensures reliable detection of faults under different conditions. The system also supports scalability, making it suitable for large-scale deployment.

Overall, the proposed system is a practical and efficient solution for modern streetlight monitoring applications. It provides a strong foundation for future enhancements and integration into smart city infrastructure.

3.9 SYSTEM PERFORMANCE ANALYSIS

The performance of the proposed smart streetlight fault detection system is evaluated based on its ability to accurately detect faults, respond in real time, and maintain system stability under different operating conditions. The system continuously monitors multiple parameters such as voltage, current, and light intensity, ensuring that any abnormal condition is detected immediately. This multi-parameter monitoring approach improves the reliability of fault detection compared to traditional systems.

One of the key performance factors is response time. The ESP32 microcontroller processes sensor data rapidly and updates the system status without delay. This ensures that faults such as lamp failure, voltage drop, or abnormal current consumption are identified almost instantly. The quick response enables maintenance teams to take corrective actions without prolonged downtime.

Accuracy is another important aspect of system performance. By analyzing both electrical and environmental parameters simultaneously, the system reduces the chances of false fault detection. For example, a streetlight may appear off during daytime conditions, but the LDR sensor ensures that such conditions are not incorrectly identified as faults. This improves overall system accuracy.

The system also demonstrates stable performance under varying load conditions. Even when multiple streetlights are connected, the centralized monitoring architecture ensures efficient data handling and processing. This makes the system reliable for both small-scale and large-scale implementations.

3.10 LIMITATIONS OF PROPOSED SYSTEM

- Although the proposed system offers significant advantages, it also has certain limitations that must be considered. One of the primary limitations is the lack of advanced predictive capabilities. The system is designed to detect faults based on real-time data, but it does not predict potential failures before they occur. Incorporating artificial intelligence techniques could further enhance the system by enabling predictive maintenance.
- Another limitation is the dependency on sensor accuracy. The performance of the system relies heavily on the proper functioning of current sensors, voltage sensors, and LDR sensors. Any error or malfunction in these sensors may lead to incorrect fault detection. Therefore, proper calibration and maintenance of sensors are essential.
- The system also depends on a stable power supply for continuous operation. In case of severe power disturbances, the system may temporarily lose monitoring capability. Although protective components are included, extreme conditions can still affect performance.
- Additionally, the communication capability of the system is limited to basic monitoring and alert generation. Advanced features such as cloud storage, remote analytics, and mobile application integration are not included in the current design. These features can be added in future enhancements to improve system functionality.

3.11 REAL-TIME IMPLEMENTATION CONSIDERATIONS

The practical implementation of the proposed system requires careful planning and consideration of real-world conditions. One of the major factors is installation. The

centralized architecture simplifies wiring and reduces the number of components required, making installation easier compared to pole-wise systems. However, proper placement of sensors and connections is essential to ensure accurate monitoring.

Environmental conditions play a crucial role in system performance. Streetlight systems are exposed to outdoor environments, including rain, dust, heat, and electrical disturbances. Therefore, all components must be properly enclosed and protected to ensure long-term reliability. Waterproof casings and insulated wiring can be used to enhance durability.

Maintenance is another important consideration. Although the system reduces manual inspection, periodic checking of sensors and connections is necessary to ensure proper operation. The system provides clear fault indications, which helps technicians quickly identify and resolve issues.

Scalability is an important advantage of the proposed system. The design can be easily expanded to include additional streetlights without major modifications. This makes the system suitable for both small installations and large urban deployments. The ESP32 microcontroller provides sufficient processing capability to handle multiple inputs efficiently.

Communication reliability is also essential for real-time monitoring. A stable network connection ensures that alerts and system updates are transmitted without delay. Any communication failure may affect the timely detection and reporting of faults. Overall, the proposed system is designed with practical implementation in mind. With proper installation, maintenance, and environmental protection, the system can operate efficiently in real-world conditions and provide reliable streetlight monitoring.

CHAPTER – 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 INTRODUCTION

The hardware and software components of the proposed smart streetlight monitoring system play a crucial role in ensuring accurate data acquisition, efficient processing, and reliable communication. The system is designed using a combination of sensors, microcontrollers, communication modules, and software tools that work together to achieve real-time fault detection. The selection of appropriate hardware components is essential to ensure system reliability, durability, and cost-effectiveness. Similarly, the choice of software tools determines the efficiency of system implementation and performance. The proposed system utilizes widely available and cost-effective components, making it suitable for large-scale deployment. The integration of hardware and software ensures smooth operation and enables continuous monitoring of streetlights under different environmental conditions. This chapter provides a detailed description of the hardware components used in the system, their working principles, and their role in the overall system operation. It also explains the software tools and programming techniques used to implement the monitoring system.

4.2 ESP32 MICROCONTROLLER

The ESP32 microcontroller serves as the central control unit of the proposed system and is responsible for processing sensor data, executing fault detection algorithms, and managing communication. It is a powerful and versatile microcontroller that offers high processing speed, low power consumption, and integrated Wi-Fi capability. The ESP32 is based on dual-core processor architecture, which allows it to handle multiple tasks simultaneously. This makes it suitable for real-time applications such as Streetlight Monitoring, where continuous data processing is required. The microcontroller includes multiple analog-to-digital converter channels, enabling it to read analog signals from sensors accurately. It also supports various communication protocols, including Wi-Fi, Bluetooth, SPI, and I2C, providing flexibility in system design. The built-in Wi-Fi module eliminates the need for external communication hardware, reducing system complexity and cost. The ESP32 is widely used in IoT applications due to its reliability and performance, making it

an ideal choice for the proposed system. Its ability to process large amounts of data quickly ensures efficient fault detection and real-time monitoring.

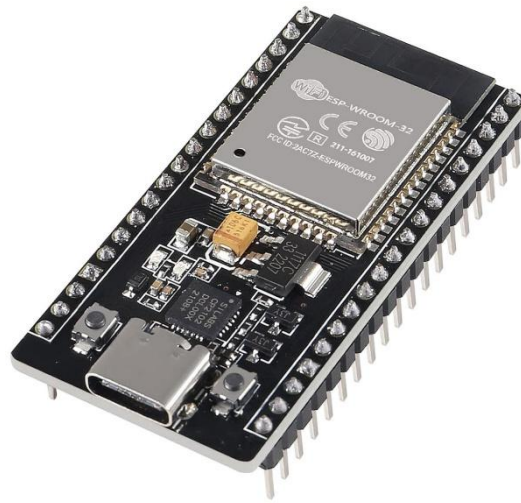


Fig 4.1: ESP-32

4.2.1 KEY FEATURES

The ESP32 microcontroller offers several features that make it suitable for the proposed system. It operates at a high clock speed, enabling fast data processing and efficient execution of embedded programs. The presence of multiple ADC channels allows it to read sensor data accurately and simultaneously. The built-in Wi-Fi capability supports wireless communication, enabling real-time data transmission without additional modules. The microcontroller also supports low power modes, which help reduce energy consumption. Its compact size and low cost make it suitable for large-scale deployment. The availability of multiple input/output pins allows easy interfacing with sensors and other components. These features make the ESP32 a reliable and efficient choice for smart streetlight monitoring systems.

4.3 CURRENT SENSOR

The current sensor is used to measure the current load flowing through the streetlight circuit and plays a crucial role in fault detection. It provides real-time information about the electrical behavior of the system, allowing the detection of abnormal conditions such as overload or open circuit. Under normal operating conditions, the current remains within a

specific range. If the current exceeds the threshold value, it indicates an overload condition, which may lead to damage of electrical components. Similarly, if the current drops significantly, it indicates a possible lamp failure or disconnection. The current sensor generates an analog output proportional to the measured current, which is then processed by the ESP32 microcontroller. Accurate measurement of current is essential for reliable fault detection, and proper calibration of the sensor ensures accurate readings. The current sensor is designed to operate under varying load conditions, making it suitable for real-world applications.

4.3.1 KEY FEATURES

The current sensor provides real-time measurement of electrical current and supports accurate fault detection. It produces an analog output compatible with the ADC of the ESP32 microcontroller. It can detect both overload and open circuit conditions. The sensor operates efficiently under varying load conditions and provides stable readings. Its compact design and ease of integration make it suitable for embedded systems.

4.4 VOLTAGE SENSOR

The voltage sensor is used to monitor the supply voltage of the streetlight system and plays an important role in detecting power-related faults. It ensures that the system receives a stable voltage required for proper operation. Any deviation from the normal voltage range can indicate faults such as power failure, line disconnection, or voltage fluctuation. The voltage sensor measures the input voltage and converts it into a proportional analog signal that can be read by the microcontroller. This allows the system to continuously monitor voltage levels and detect abnormal conditions in real time. The sensor is designed to handle different voltage ranges and provide accurate readings. Proper voltage monitoring improves the reliability of the system and prevents damage to electrical components.

4.5 LDR SENSOR

The Light Dependent Resistor (LDR) sensor is used to measure the brightness level of the streetlight and confirm whether the lamp is functioning properly. The resistance of the LDR changes based on the intensity of light falling on it. In bright conditions, the

resistance decreases, while in dark conditions, the resistance increases. This property allows the LDR to detect whether the streetlight is emitting light. The sensor generates a voltage signal corresponding to the light intensity, which is read by the microcontroller. By analyzing the LDR readings along with voltage and current values, the system can accurately determine whether the streetlight is operating correctly. The LDR sensor is simple, cost-effective, and easy to integrate, making it suitable for the proposed system.



Fig 4.2: LDR Sensor

4.5.1 KEY FEATURES

The LDR sensor detects light intensity and confirms the ON/OFF status of the streetlight. It provides analog output compatible with the microcontroller. It is low cost and easy to integrate into the system. The sensor operates reliably under different lighting conditions and supports accurate fault detection.

4.6 COMMUNICATION MODULE (WI-FI)

The communication module enables the system to transmit data and alerts to a remote monitoring platform. The ESP32 microcontroller includes a built-in Wi-Fi module, which allows wireless communication without additional hardware. This feature simplifies system design and reduces cost. The Wi-Fi module enables real-time data transmission, ensuring that fault information is communicated instantly. The system can be connected to a local network or the internet, allowing remote monitoring and control. The use of Wi-Fi provides high data transmission speed and supports integration with cloud platforms. Reliable

communication is essential for the effectiveness of the system, as it ensures timely alert generation and response.

4.6.1 KEY FEATURES

The Wi-Fi module supports real-time data transmission and remote monitoring. It eliminates the need for external communication devices, reducing system complexity. It provides high-speed communication and supports integration with cloud platforms. The module is energy-efficient and suitable for IoT applications.

4.7 POWER SUPPLY UNIT

The power supply unit provides the necessary voltage required for the operation of the microcontroller and sensors. It converts the input voltage into a stable and regulated output suitable for electronic components. The ESP32 operates at 3.3V, while some sensors may require 5V, making voltage regulation essential. The power supply unit includes components such as transformers, rectifiers, and voltage regulators to ensure stable output. Proper filtering is used to eliminate noise and fluctuations in the power supply. A stable power supply is critical for accurate sensor readings and reliable system operation. Any fluctuation in voltage can affect the performance of the system and lead to incorrect fault detection.

4.8 EMBEDDED C PROGRAMMING

Embedded C is used to program the ESP32 microcontroller and implement the fault detection logic. The program continuously reads sensor data, processes it, and compares it with predefined threshold values. Based on the analysis, the system determines whether the streetlight is functioning normally or experiencing a fault. The program also handles communication by sending alerts through the Wi-Fi module. Embedded C is preferred for its efficiency, speed, and control over hardware. It allows precise implementation of real-time applications and ensures reliable system performance.

4.9 ARDUINO IDE

The Arduino IDE is used as the development environment for writing, compiling, and uploading the program to the ESP32 microcontroller. It provides a user-friendly interface and supports various libraries required for sensor interfacing and communication. The IDE allows easy debugging and testing of the system using the serial monitor. It simplifies the development process and enables efficient implementation of the monitoring system. The availability of libraries and community support makes Arduino IDE a suitable choice for developing embedded systems.

4.10 SENSOR INTERFACING AND CALIBRATION

Sensor interfacing is a critical aspect of the proposed system, as it ensures accurate data acquisition from the streetlight network. The sensors used in the system, including current sensor, voltage sensor, and LDR sensors, generate analog signals that must be properly interfaced with the ESP32 microcontroller. These signals are connected to the analog input pins of the microcontroller, where they are converted into digital values using the built-in analog-to-digital converter. Proper interfacing requires the use of signal conditioning techniques such as voltage division, filtering, and amplification to ensure compatibility with the microcontroller's input range. Calibration of sensors is equally important to achieve accurate measurements. Each sensor must be calibrated to ensure that the output values correspond correctly to the actual physical parameters. For example, the current sensor must be calibrated to accurately measure the load current under different conditions, while the voltage sensor must be adjusted to provide precise voltage readings. The LDR sensor must be calibrated to distinguish between normal light levels and faulty conditions. Improper calibration can lead to incorrect fault detection, reducing system reliability. Therefore, careful calibration and testing of sensors are essential steps in system development.

4.11 HARDWARE RELIABILITY AND DURABILITY

The reliability and durability of hardware components are crucial for the long-term operation of the streetlight monitoring system. Since the system is deployed in outdoor environments, it is exposed to various environmental conditions such as temperature variations, humidity, dust, and electrical disturbances. These factors can affect the performance of electronic components and lead to system failure if not properly addressed.

The proposed system is designed to ensure high reliability by using robust components and proper protective measures. The sensors and microcontroller are enclosed in protective casings to prevent damage from environmental exposure. Proper insulation is used to protect the circuitry from moisture and dust. Additionally, the system is designed to handle voltage fluctuations and electrical noise, which are common in outdoor power systems. The use of voltage regulators and filtering circuits helps maintain stable operation and prevents damage to sensitive components. Regular maintenance and periodic inspection can further improve system durability. By ensuring reliable hardware design, the system can operate continuously for long periods without failure.

4.12 SOFTWARE DESIGN AND CONTROL LOGIC

The software design of the proposed system plays a vital role in its overall functionality and performance. The embedded program developed using Embedded C is responsible for controlling all system operations, including data acquisition, processing, fault detection, and communication. The program is designed to operate in a continuous loop, where sensor data is repeatedly collected and analyzed. The control logic includes predefined threshold values for voltage, current, and light intensity, which are used to determine the normal operating conditions. When the sensor readings deviate from these thresholds, the program identifies a fault condition and determines the type of fault based on the combination of parameter values. The software also includes communication routines that enable the system to send alerts through the Wi-Fi module. Error handling mechanisms are incorporated to ensure stable operation even in the presence of noise or communication failures. The program is optimized for efficient execution, ensuring fast response time and minimal delay in fault detection. Proper software design ensures accurate and reliable system performance.

4.13 SYSTEM TESTING AND VALIDATION

System testing and validation are essential steps in ensuring the proper functioning of the proposed streetlight monitoring system. After integrating all hardware and software components, the system is tested under different operating conditions to verify its performance. Testing involves checking the accuracy of sensor readings, verifying the fault detection logic, and ensuring proper communication of alerts. Different test cases are conducted, including normal operation, lamp failure, overload condition, and power failure. During testing, the system's response to each condition is observed and analyzed. The

accuracy of fault detection is evaluated by comparing sensor readings with expected values. Any discrepancies are corrected through calibration and software adjustments. The communication system is also tested to ensure that alerts are transmitted correctly and without delay. Validation of the system confirms that it meets the required performance criteria and operates reliably under real-world conditions. Proper testing ensures that the system is ready for practical deployment.

CHAPTER - 5

SIMULATION RESULTS AND DISCUSSIONS

5.1 INTRODUCTION

The simulation of the proposed smart streetlight monitoring system was carried out to evaluate the overall performance, accuracy, and reliability of the system under different operating conditions. The main objective of this simulation is to verify the ability of the system to detect faults in streetlights using real-time monitoring of electrical parameters such as voltage, current, and light intensity. The simulation also helps in understanding how effectively the system can distinguish between normal working conditions and faulty conditions.

The system is designed using a centralized row-wise monitoring approach, where multiple streetlights are monitored using a single microcontroller unit. This reduces hardware complexity and improves efficiency compared to traditional pole-wise monitoring systems. The simulation environment replicates real-world conditions, allowing detailed observation of system behavior without the need for physical implementation.

5.2 SIMULATION SETUP

The simulation setup consists of an ESP32 microcontroller interfaced with multiple sensors and LED loads representing streetlights. Each streetlight is connected with a current sensor, voltage sensor, and LDR sensor to monitor its operational status. The ESP32 collects data from all sensors and processes it to determine whether each streetlight is functioning properly.

The LDR sensor is used to detect ambient light conditions, enabling automatic identification of day and night scenarios. The current sensor measures the flow of current through each streetlight, while the voltage sensor monitors the supply voltage. These parameters are continuously analyzed to detect abnormal conditions such as lamp failure, low current, or voltage fluctuations. The system is designed in such a way that all components are interconnected to ensure smooth data acquisition and processing. The simulation also includes a serial monitor interface, which displays real-time data for analysis

and verification.

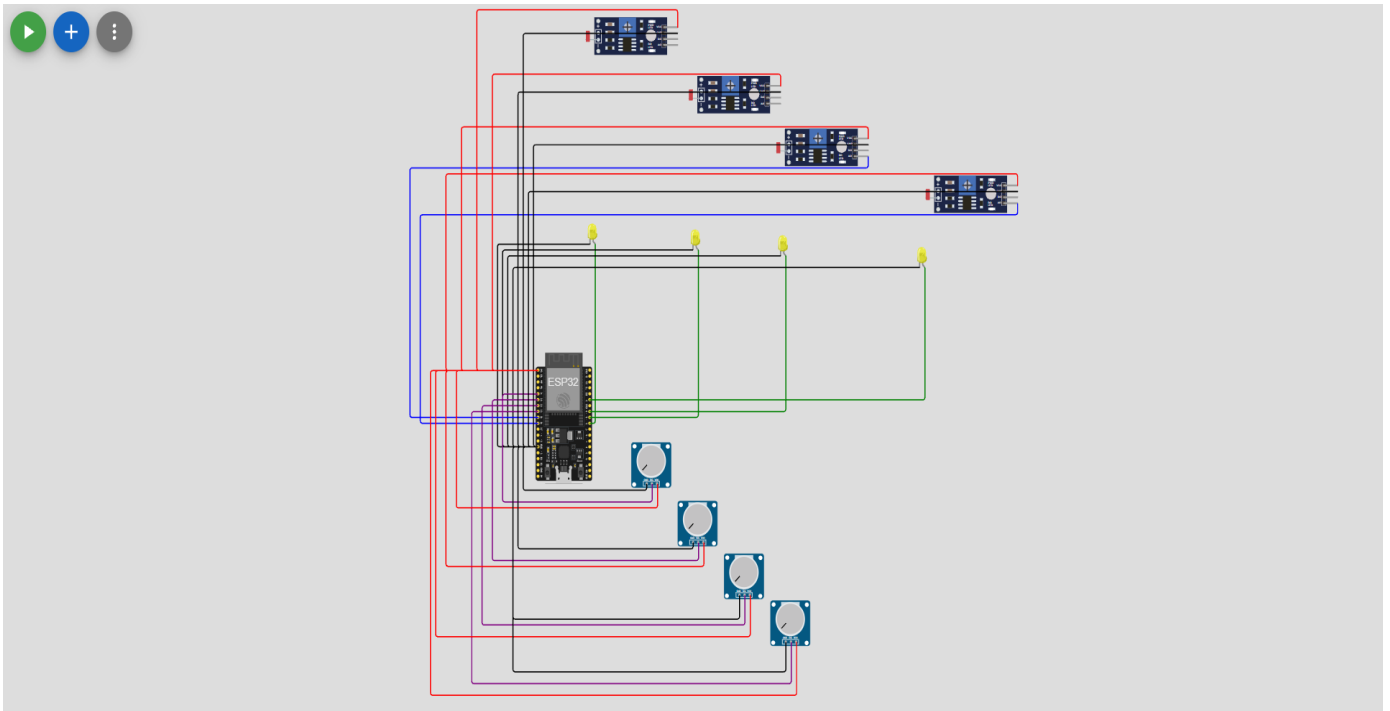


Fig. 5.1 Simulation Circuit Diagram of Smart Streetlight System

5.3 WORKING OF SIMULATION

During the simulation, the system continuously reads input values from all sensors and processes them using predefined logic. The LDR sensor determines whether it is day or night and accordingly activates or deactivates the streetlights. When the LDR detects low light conditions, the system turns ON the streetlights and begins monitoring their performance.

The voltage and current values are continuously monitored to ensure proper operation. If the current and voltage remain within acceptable limits and the LDR confirms light emission, the system marks the streetlight status as “WORKING.” However, if any parameter falls outside the threshold range, the system identifies it as a fault condition.

The ESP32 microcontroller processes all sensor data in real time and updates the status of each streetlight. The results are displayed on the serial monitor, allowing easy observation of system performance.

5.4 SIMULATION RESULTS

The simulation results demonstrate the effectiveness of the proposed system in detecting faults accurately. The serial monitor output displays real-time values of voltage, current, and LDR readings for each streetlight. Based on these values, the system determines whether the streetlight is functioning normally or experiencing a fault.

In the observed results, some streetlights are marked as “WORKING,” while others are marked as “FAULT.” This indicates that the system can independently monitor multiple streetlights and accurately identify faulty units. For example, when the LDR value is low and current is insufficient, the system correctly identifies the streetlight as faulty. The results also show that the system responds quickly to changes in sensor values. When a fault condition occurs, the status is updated immediately, ensuring timely detection and response.

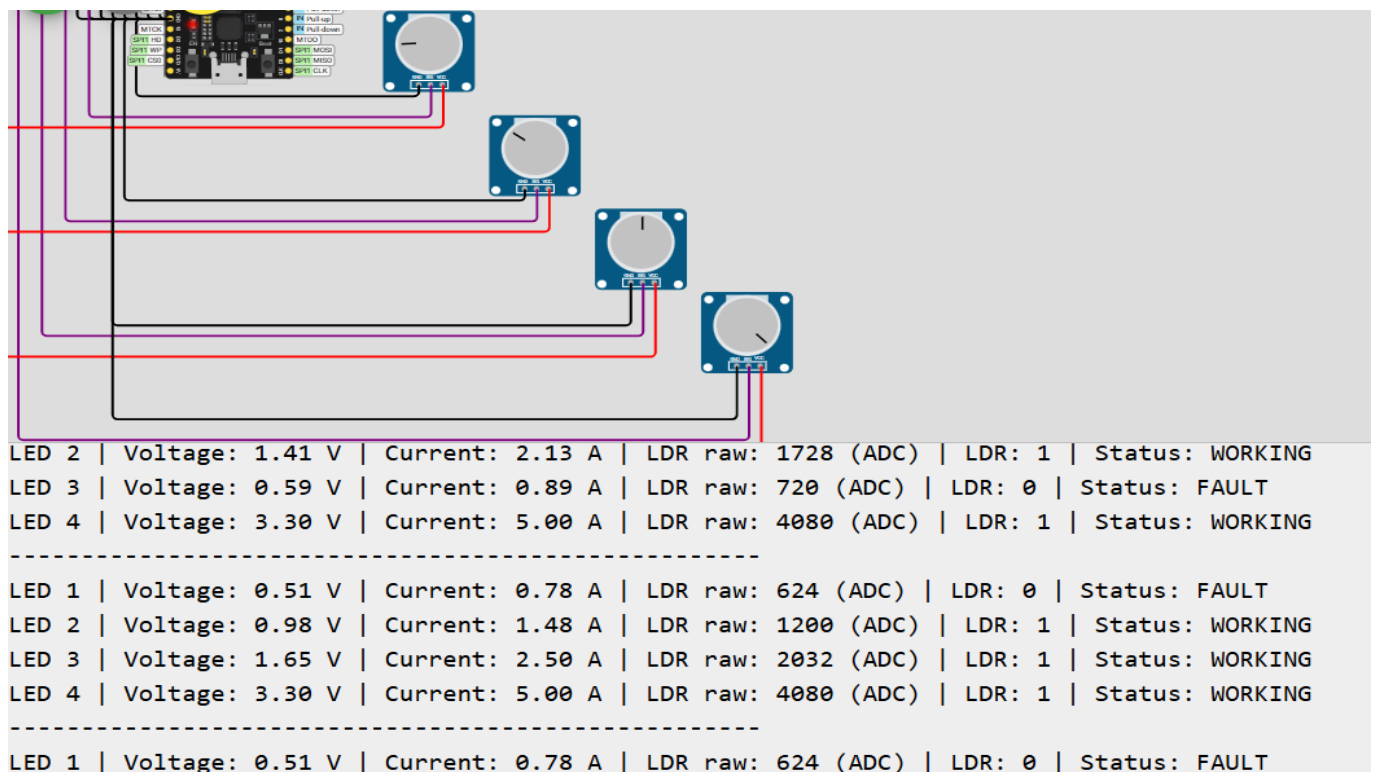


Fig. 5.2 Serial Monitor Output

5.5 PERFORMANCE ANALYSIS

The performance of the system was evaluated based on accuracy, response time, and reliability. The simulation results indicate that the system achieves high accuracy in detecting faults by analyzing multiple parameters simultaneously. This multi-parameter approach reduces the chances of false detection and improves overall system performance.

The response time of the system is very low, as faults are detected almost instantly after they occur. This is essential for real-world applications, where quick detection can reduce maintenance time and improve safety. The centralized monitoring approach also enhances efficiency by reducing hardware requirements and simplifying system design.

The system performs consistently under different operating conditions, indicating its robustness and reliability. The simulation confirms that the proposed system is suitable for large-scale deployment.

5.6 FAULT DETECTION ANALYSIS

The fault detection mechanism is based on comparing sensor values with predefined threshold levels. When the current value drops below a certain limit or when the LDR fails to detect light, the system identifies it as a fault. Similarly, abnormal voltage levels also indicate potential issues in the system.

The simulation demonstrates that the system can detect different types of faults, including lamp failure, low current conditions, and voltage fluctuations. Each fault condition is clearly identified and displayed on the serial monitor.

The ability to detect faults at an individual streetlight level makes the system highly effective and efficient. This reduces the need for manual inspection and enables faster maintenance.

5.7 ADVANTAGES OBSERVED IN SIMULATION

The simulation highlights several advantages of the proposed system. Centralized architecture reduces hardware complexity and cost, making the system more economical.

Real-time monitoring capability ensures continuous observation of streetlight performance, improving reliability.

The system also provides accurate fault detection using multiple sensor inputs, reducing the chances of error. The use of ESP32 microcontroller enables fast data processing and efficient communication. Overall, the simulation confirms that the system is efficient, reliable, and scalable.

5.8 LIMITATIONS OF SIMULATION

Despite the successful results, certain limitations were observed during the simulation. The simulation environment may not fully represent real-world conditions such as environmental disturbances, electrical noise, and communication delays. These factors may affect system performance in practical applications.

The system currently uses fixed threshold values for fault detection, which may need adjustment for different environments. Additionally, the system does not include predictive capabilities, which could further enhance performance.

5.9 DISCUSSION

The simulation results clearly validate the effectiveness of the proposed smart streetlight monitoring system. The system successfully detects faults in real time and provides accurate status updates for each streetlight. The use of multiple sensors ensures reliable fault detection, while the centralized architecture reduces complexity and cost.

The results demonstrate that the system is capable of handling multiple streetlights simultaneously without compromising performance. This makes it suitable for large-scale deployment in urban and rural areas.

Overall, the simulation confirms that the proposed system is a practical and efficient solution for smart streetlight monitoring. With further enhancements, the system can be integrated into smart city infrastructure to improve energy efficiency and maintenance management.

CHAPTER - 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The proposed Smart Streetlight Fault Detection and Monitoring System successfully demonstrates an efficient and reliable approach to streetlight management using modern embedded and sensing technologies. The system is designed to continuously monitor multiple streetlights using a centralized row-wise architecture, significantly reducing hardware complexity and overall system cost when compared to traditional pole-wise monitoring systems.

The integration of sensors such as current sensors, voltage sensors, and LDR sensors enables accurate real-time monitoring of streetlight performance. By analyzing these parameters, the system effectively identifies various fault conditions, including lamp failure, abnormal current flow, and voltage fluctuations. The use of the ESP32 microcontroller ensures fast data processing and efficient handling of multiple inputs simultaneously.

One of the key achievements of this system is its ability to provide immediate fault detection and status updates. The simulation results confirm that the system can clearly distinguish between working and faulty streetlights with high accuracy. This reduces the need for manual inspection and enables faster maintenance, ultimately improving system reliability and reducing operational costs.

The centralized monitoring approach also enhances scalability, allowing the system to be easily expanded for larger streetlight networks. The design ensures minimal energy wastage by identifying faulty lights that consume power without providing proper illumination. This contributes to energy efficiency and supports sustainable infrastructure development.

Overall, the proposed system proves to be a practical, cost-effective, and scalable solution for modern streetlight monitoring applications. It addresses the limitations of existing systems and provides a strong foundation for further advancements in smart city technologies.

6.2 FUTURE SCOPE

Although the proposed system performs effectively in fault detection and monitoring, there are several opportunities for further enhancement and development. Future improvements can focus on increasing system intelligence, expanding functionality, and improving real-world adaptability.

One of the major enhancements is the integration of cloud-based monitoring systems. By connecting the system to cloud platforms, real-time data can be stored, analyzed, and accessed remotely. This enables centralized control and monitoring of streetlight networks from any location, improving efficiency and convenience.

Another important improvement is the incorporation of artificial intelligence and machine learning techniques. These technologies can be used to analyze historical data and predict potential failures before they occur. Predictive maintenance can significantly reduce downtime and improve system reliability.

The system can also be enhanced by developing a mobile or web-based application interface. This would allow maintenance personnel to receive instant notifications, monitor system status, and manage streetlights more efficiently. Such integration would make the system more user-friendly and accessible.

Future work may also include the integration of renewable energy sources such as solar panels. Solar-powered streetlights combined with smart monitoring can further reduce energy consumption and promote environmentally sustainable solutions.

Additionally, advanced communication technologies such as GSM, LoRa, or 5G can be incorporated to improve data transmission reliability, especially in remote or large-scale deployments. This would ensure uninterrupted communication between the monitoring system and control units.

In conclusion, the proposed system has strong potential for future development and integration into smart city infrastructure. With the addition of advanced technologies and intelligent features, it can evolve into a fully automated, energy-efficient, and highly reliable streetlight management system.

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- [6] Smart City Lighting Infrastructure – Research highlighting scalable and cost-effective smart lighting deployment strategies.
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- [20] IoT-Based Smart Street light Monitoring System –
Research on real-time monitoring and automated fault detection of streetlights using IoT-enabled devices.
- [21] Intelligent Street Lighting Using Embedded Systems – Studies on automated control and fault identification in street lighting using microcontrollers and sensors.
- [22] Smart LED Street Lighting with Energy Optimization – Research focusing on adaptive brightness control and energy-efficient LED lighting systems.

[23] IoT-Based Fault Detection in Urban Lighting Systems – Approaches for detecting and reporting streetlight failures using wireless communication and IoT platforms.

[24] Wireless Sensor Network-Based Smart Lighting Systems – Studies on distributed sensor networks for monitoring, controlling, and maintaining streetlight infrastructure.

[25] Smart Streetlight Fault Detection and Maintenance System – Research on automated fault identification and maintenance optimization in street lighting using sensor-based monitoring systems.

APPENDIX

SOURCE CODE

```
#include <WiFi.h>
#include <HttpClient.h>

const char* ssid = "YOUR_WIFI_SSID";
const char* password = "YOUR_WIFI_PASSWORD";
const char* serverUrl = "http://192.168.1.100:5000/update";

#define LDR1 34
#define LDR2 35
#define LDR3 32
#define CURRENT_SENSOR 33

#define ADC_MAX 4095.0
#define VREF 3.3

float zeroOffset = 0;

float getVoltage(int raw) {
    return (raw / ADC_MAX) * VREF;
}

String getStatus(float v) {
    if (v < 1.5) return "working";
    else return "not_working";
}

void calibrateCurrent() {
    long sum = 0;
    for (int i = 0; i < 200; i++) {
        sum += analogRead(CURRENT_SENSOR);
        delay(2);
    }
    zeroOffset = sum / 200.0;
}

float getCurrent() {
    long sum = 0;
    for (int i = 0; i < 100; i++) {
        sum += analogRead(CURRENT_SENSOR);
    }
    float avg = sum / 100.0;
    float value = avg - zeroOffset;
    if (value < 0) value = 0;
    return value * 0.01;
}
```

```

void sendData(float current, String s1, String s2, String s3) {
    if (WiFi.status() != WL_CONNECTED) return;

    HTTPClient http;
    http.begin(serverUrl);
    http.addHeader("Content-Type", "application/json");

    String body = "{}";
    body += "\"current\": " + String(current, 2) + ",";
    body += "\"light1\": \"" + s1 + "\",";
    body += "\"light2\": \"" + s2 + "\",";
    body += "\"light3\": \"" + s3 + "\"";
    body += "}";

    int httpCode = http.POST(body);
    Serial.println(httpCode);
    http.end();
}

void setup() {
    Serial.begin(115200);

    pinMode(LDR1, INPUT);
    pinMode(LDR2, INPUT);
    pinMode(LDR3, INPUT);
    pinMode(CURRENT_SENSOR, INPUT);

    WiFi.begin(ssid, password);
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
    }

    calibrateCurrent();
}

void loop() {
    int r1 = analogRead(LDR1);
    int r2 = analogRead(LDR2);
    int r3 = analogRead(LDR3);

    float v1 = getVoltage(r1);
    float v2 = getVoltage(r2);
    float v3 = getVoltage(r3);

    String s1 = getStatus(v1);
    String s2 = getStatus(v2);
    String s3 = getStatus(v3);
}

```

```
float current = getCurrent();  
  
sendData(current, s1, s2, s3);  
  
delay(1000);  
}
```

CERTIFICATE

OF PARTICIPATION

INTERNATIONAL CONFERENCE ON SEMICONDUCTOR, SMART GRID & INTELLIGENT COMMUNICATION TECHNOLOGIES

ICSSICT'26

This is to certify that Dr./Mr./Ms. PREM ARVINTH. G.

SRI VENKATESWARA COLLEGE OF TECH, AUTUMATED SMART STREET LIGHT FAULT DETECTION AND RESPONSE
has participated in the International Conference on Semiconductor, Smart Grid & Intelligent
Communication Technologies (**ICSSICT'26**) organized by the Department of ECE & EEE, PERI Institute
of Technology, Chennai, held on 13th April 2026.

Rajenar
HOD/ECE

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HOD/EEE

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VICE PRINCIPAL

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Rajaguru
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We sincerely appreciate their creativity, innovation, and active involvement in presenting their project.


Mr. M. Prabakaran
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